

Sahand Rahmdel

EDUCATION

M.Sc. Computer Science, Specialization: Artificial Intelligence and Data RWTH Aachen University	Aachen, Germany Apr. 2025 – Est. Mar. 2027
<i>Relevant Coursework:</i> Machine Learning, Advanced ML, Data Science, Reinforcement Learning, Computer Vision, Business Process Intelligence, Actions & Planning in AI, Quantum Computing <i>Seminar Neuroinspired Computing – Reinforcement Learning</i> [Link]: A paper on computational framework and biological foundations of Reinforcement Learning, Grade 1.0.	
B.Sc. Computer Science including Coursework in Business Administration RWTH Aachen University	Aachen, Germany Nov. 2020 – Mar. 2025
<i>Seminar Neuroinspired Computing – Attention Mechanism and its Applications</i> [Link]: A paper on biological vs. computational Attention Mechanism; presented a deep-dive on the Transformer, Grade 1.3.	

WORK EXPERIENCE

AI Engineer (Working Student) Mercedes-Benz AG	Nov. 2025 – Present Sindelfingen, Germany - Remote
- Sole engineer and technical lead contact of NEXA (NX Engineering eXpert Agent), an enterprise AI agentic companion; presented live architecture to 800+ attendees (500 in-person) at company town hall. - Architected NEXA's multi-agent LangGraph pipeline with local RAG (semantic and lexical search tools with reranking) and custom MCP services to automate code generation across a 70k+ LOC enterprise C# repository. - Engineered core modules for TQA, a Siemens NX CAD conformance checker; optimized unused-object analyzer to improve speed by 47%.	
Research Assistant – HPC for NLP Jülich Supercomputing Centre (JSC), Forschungszentrum Jülich	Apr. 2024 – Dec. 2024 Jülich, Germany
Built HPC evaluation pipelines to construct a multilingual embedding proximity map across 24 languages for Teuken-7B (Germany's 1st LLM), optimizing cross-lingual transfer learning and data selection.	

PROJECTS

HackerRank LLM Orchestration Hackathon Top 4% Worldwide [Link]	Jun. 2026
Built a multi-modal VLM claim reviewer with deterministic guardrails for business logic, ranking 2nd in Germany.	
Software Project Lab – Artifaix (AI-supported Information Foraging) [Link]	Oct. 2025 – Apr. 2026
Research Group Software Construction (SWC), RWTH Aachen	Aachen, Germany
- Led a 4-person team to design a modular RAG-based microservice architecture for automated knowledge extraction. - Implemented LLM reasoning and MapReduce processing for papers exceeding standard model context windows.	
Bachelor's Thesis – Exploring Linguistic Proximity [Link]	Jun. 2024 – Dec. 2024
Visual Computing Institute, RWTH Aachen & Forschungszentrum Jülich & WestAI	Aachen, Germany
Analyzed 6.1 TB of multilingual text data from 24 languages, applied t-SNE to map high-dimensional clusters discovering that raw embeddings cluster by topic, and mean embeddings cluster by linguistic family.	
Software Project Lab – Network Engineering Practical Internet eXperience (PIX)	Oct. 2022 – Apr. 2023
Chair of Communication and Distributed Systems (ComSys), RWTH Aachen	Aachen, Germany
Configured OSPF, BGP, and RPKI protocols in a simulated network of 28 virtual routers.	
System Programming Lab – Embedded Systems Programming	Oct. 2021 – Apr. 2022
Chair of Embedded Software, RWTH Aachen	Aachen, Germany
Developed low-level C firmware for ATmega644 embedded system, managing memory and peripherals (GPIO, Timers).	

TECHNICAL SKILLS

Programming: Python, C# (.NET), Java, C/C++ , Bash, TypeScript, HTML/CSS
Machine Learning & NLP: PyTorch, Hugging Face (Transformers, Datasets), Sentence Transformers, scikit-learn, Vector Embeddings, RAG, LangChain, LangGraph, ChromaDB, MCP
Data & Process Mining: pandas, NumPy, Data Visualization, ProM, Celonis
Backend & Systems: FastAPI, Microservices, LINQ, async/await, Linux, Docker, SLURM, MapReduce, Vector Databases
Databases & Infrastructure: MySQL, SQL, Git, Conda/venv/uv
Frontend & Collaboration: React, Vite, Tailwind CSS, shadcn/ui, Jira, Confluence
Engineering Tools: ETL X, RapidMiner, Siemens NX, NX Open API, Visual Studio

LANGUAGES

German: DSH-3 (C2 / Fluent)	English: Fluent	Persian: Native	Azerbaijani: Native	Turkish: Fluent
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